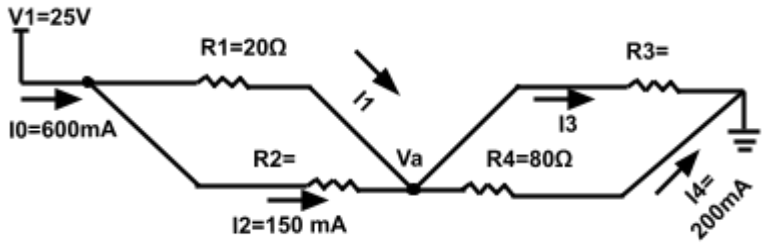
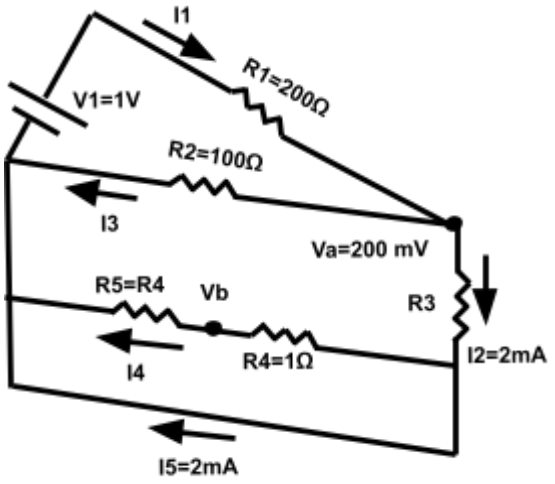
En todos los ejercicios se deben encontrar los valores faltantes, a menos que se indique lo contrario
Estos ejercicios deben ser verificados con el simulador para validar los resultados encontrados. En todo caso donde exista una única fuente, las tensiones están referidas al negativo de la misma (masa). Cuando existen dos fuentes, y los negativos de ambas se encuentran unidas, esta es la masa común de referencia.
No respondemos consultas sobre ejercicios que pueden verificarse en el simulador.

1)



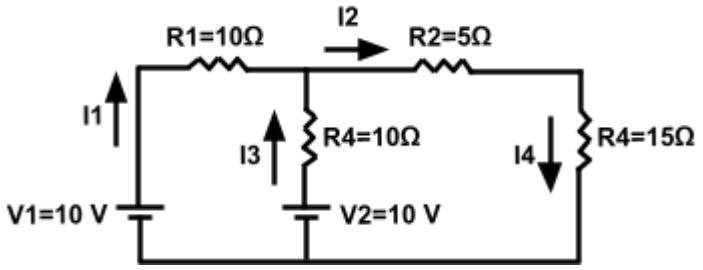
R2= **R3=** **I1=** **I3=** **Va=**

2)



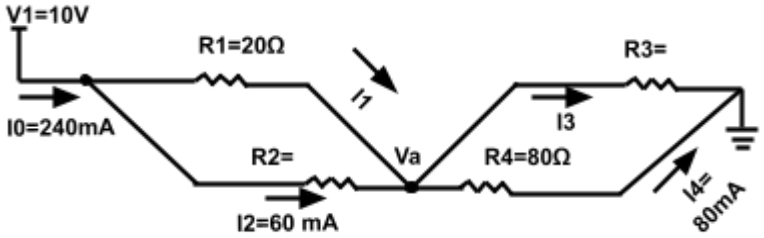
R3= **I1=** **I3=** **I4=** **Vb=**

3)



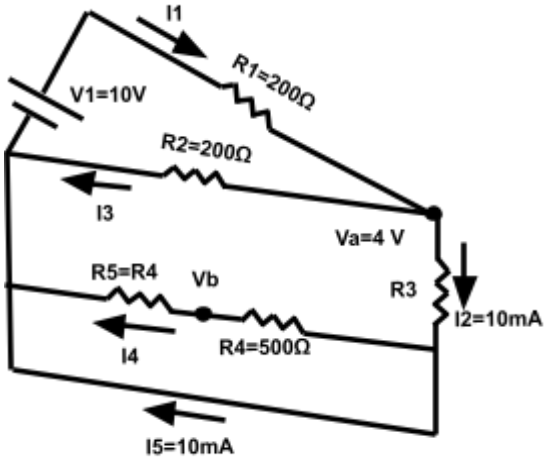
I1= **I2=** **i3=** **I4=**

4)



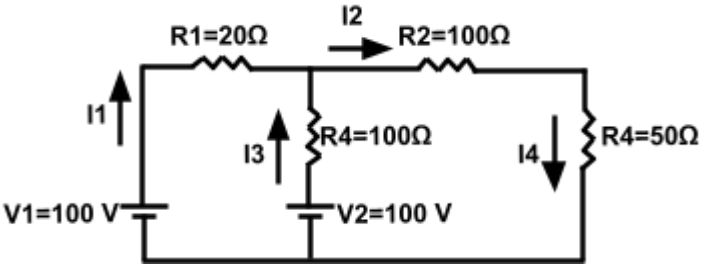
R2= **R3=** **I1=** **I3=** **Va=**

5)



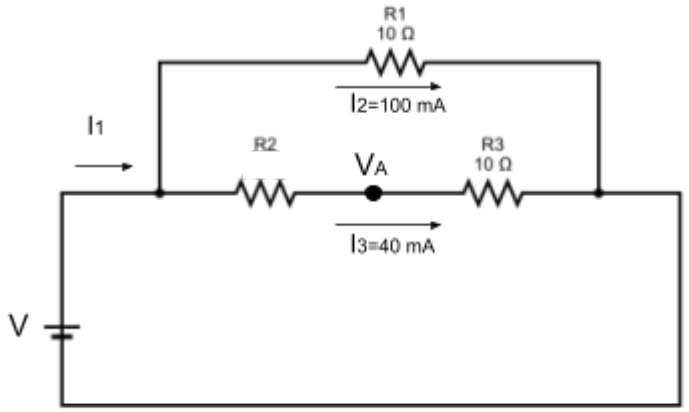
R3= **I1=** **I3=** **I4=** **Vb=**

6)



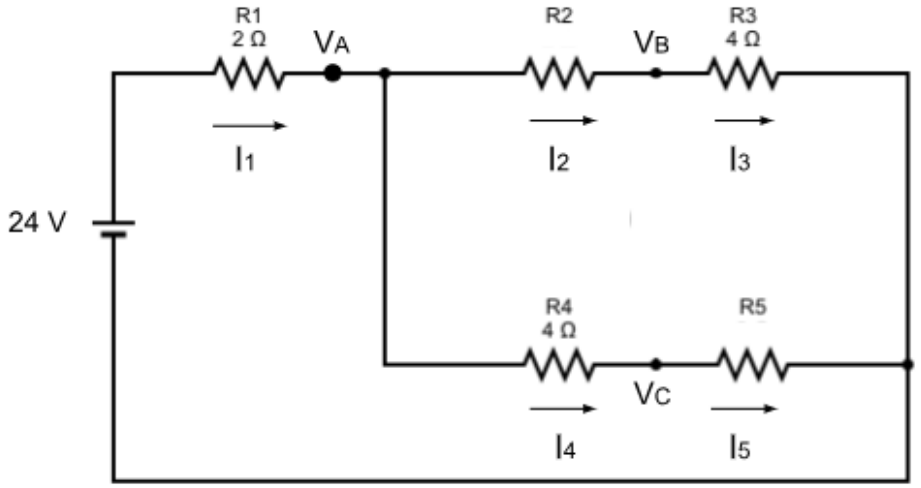
I1= **I2=** **i3=** **I4=**

7)

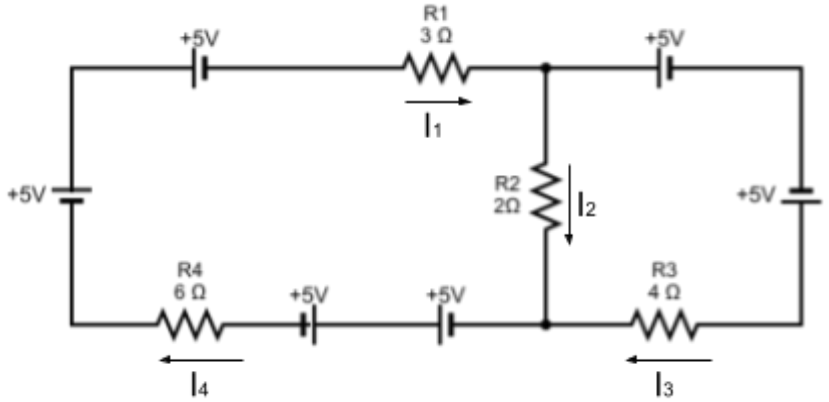


V= **I1=** **R2=** **VA=**

8) Dado el siguiente circuito, se pide hallar las corrientes que circulan por todas las resistencias y los valores de R2 y R5. Se sabe que la diferencia de potencial en VA respecto a masa es de 14.4 V. Sumado a lo anterior , también se sabe que la diferencia de potencial entre entre VB y VC es 4.8V y que el punto VC respecto a masa también es 4.8 V.

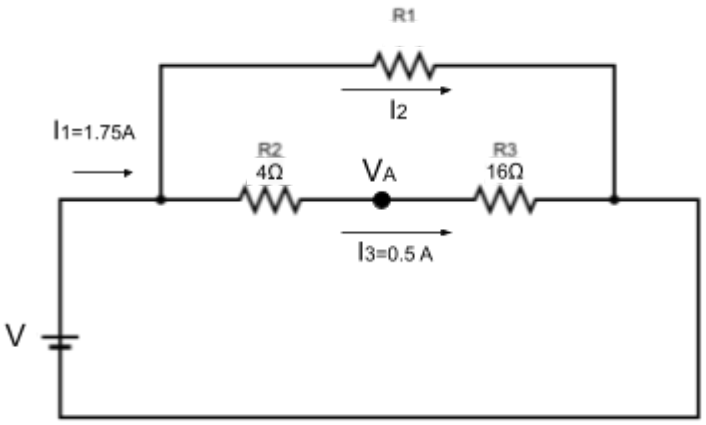


9)



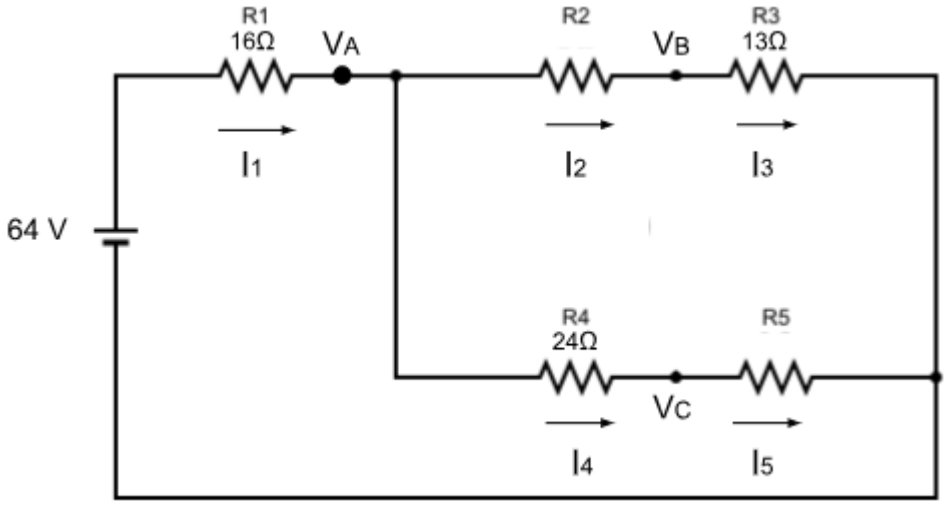
I1= **I2=** **I3=** **I4=**

10)



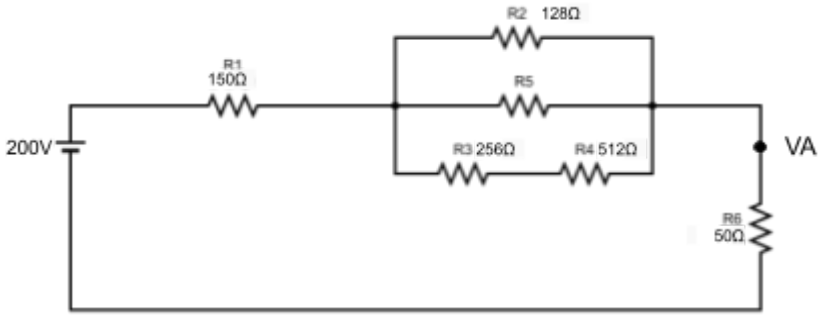
V= **R1=** **I2=** **VA**

11) Dado el siguiente circuito, se pide hallar las corrientes que circulan por todas las resistencias y los valores de R2 y R5. Se sabe que la diferencia de potencial en VA respecto a masa es de 32V . Sumado a lo anterior , también se sabe que la diferencia de potencial entre entre VB y VC es 5V y que el punto VC respecto a masa es 8 V.



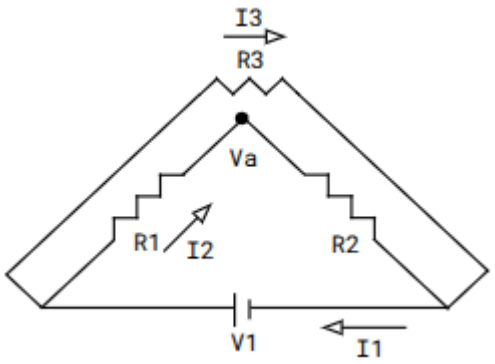
I1= **I2=** **I3=** **I4=** **I5=** **R2=** **R5=**

12) En el punto VA se sabe que la tensión respecto a masa es de 50 V . Halle la resistencia R5 y la resistencia equivalente total del circuito.



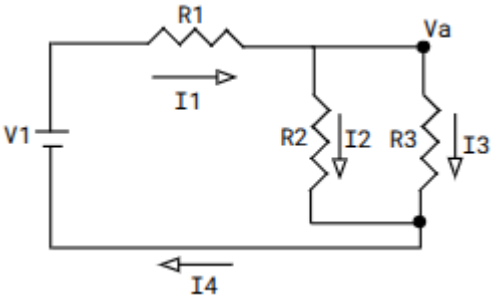
R5= **Resistencia equivalente total=**

13)



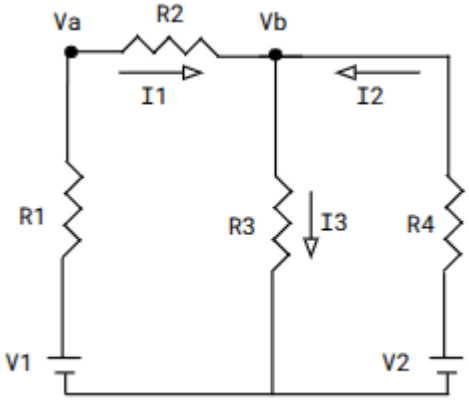
I1	132 mA
I2	
I3	120 mA
R1	
R2	750Ω
R3	
Va	
V1	12V

14)



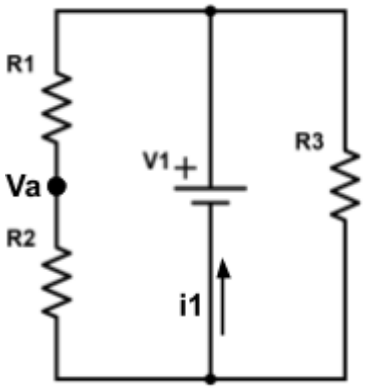
I1	125 mA
I2	
I3	
I4	
R1	
R2	400Ω
R3	
Va	40V
V1	50V

15)

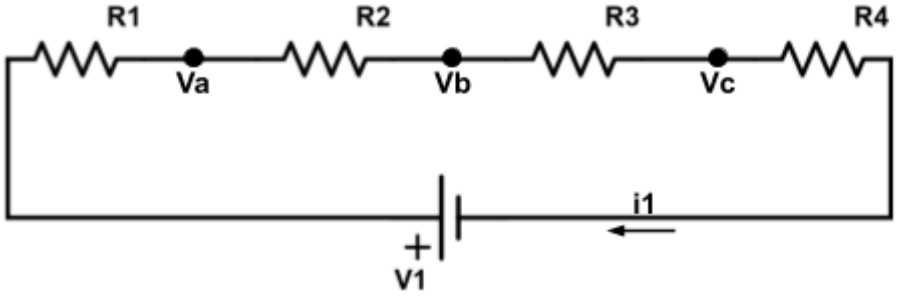


I1	
I2	
I3	
R1	50Ω
R2	50Ω
R3	400Ω
R4	100Ω
V1	60V
V2	30V
Va	
Vb	

16) Dado el siguiente circuito , se sabe que V1=100V , R1=45 Ω , R2=55Ω y R3= 50Ω
Encuentre Va e I1

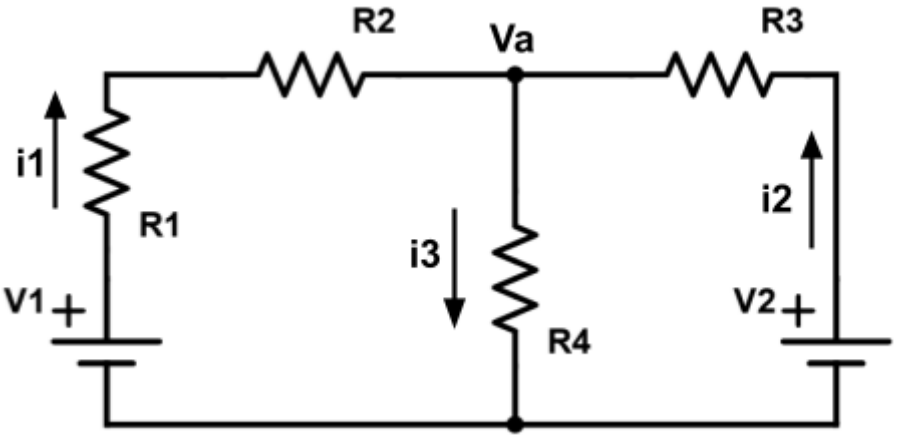


17) Dado el siguiente circuito, se sabe que R4=6Ω Va = 25V , Vb=16,5V, Vc=1,5V y V1=50V. Indique los valores faltantes

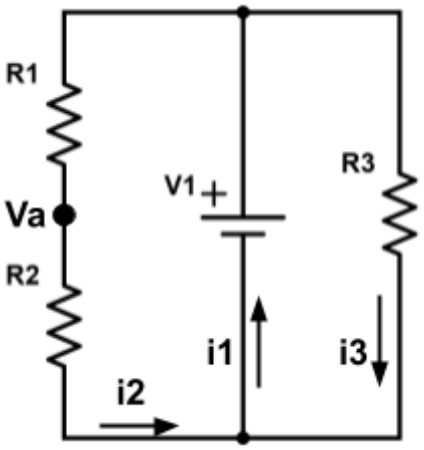


R1= **R2=** **R3=** **R4=** **I1=**

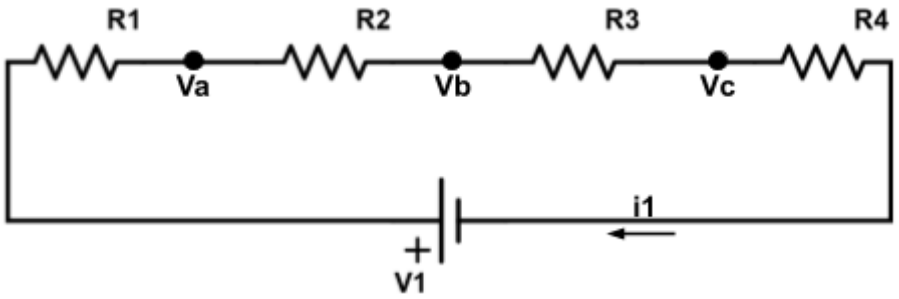
18) Dado el siguiente circuito, encuentre los valores de **i1, i2 e i3,** y la tensión en **Va.** Todas las resistencias son de **50 Ω**. **V1=20V y V2=40V**



19) Dado el siguiente circuito , se sabe que V1=100V , Va=25V , R3 = 200 Ω, i1=2A Encuentre I2 , R1 y R2

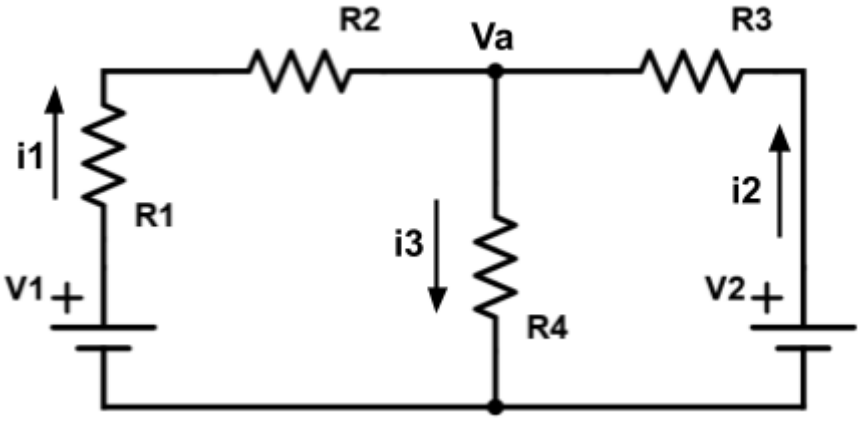


20) Dado el siguiente circuito, se sabe que R4=6Ω Va = 50V , Vb=33V, Vc=3V y V1=100V. Indique los valores faltantes:



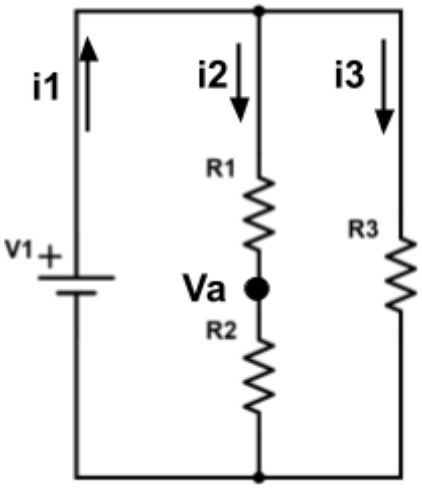
R1= **R2=** **R3=** **R4=** **I1=**

21)

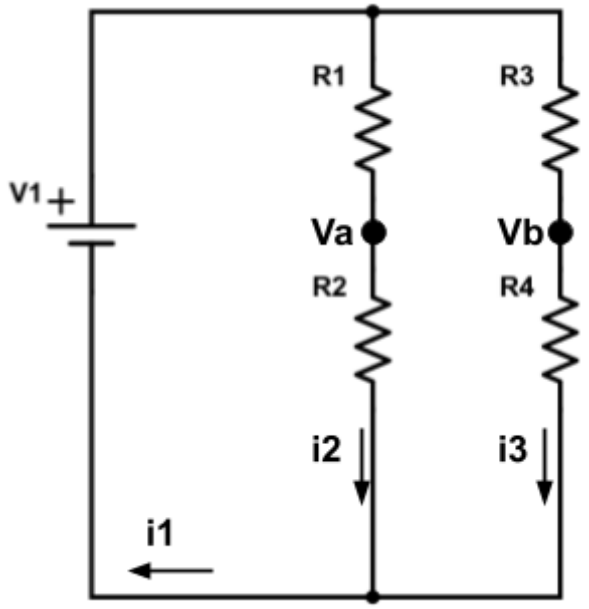


I1= **I2=** **I3=** **Va=**

22) Dado el siguiente circuito , se sabe que V1=200V , R1=45 Ω , R2=55Ω y R3= 50Ω Encuentre Va e I1

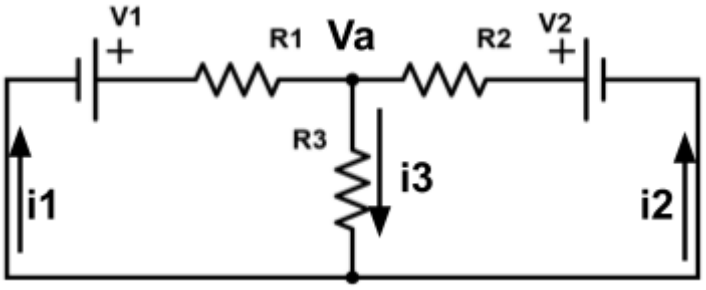


23) Dado el siguiente circuito, se sabe V1=15V, R1=0 Ω , R2=5 Ω , R3=9 Ω , R4=6 Ω. Calcule los valores de **i1, i2 e i3,** y las tensiones en **Va y Vb** con respecto al negativo de la fuente.



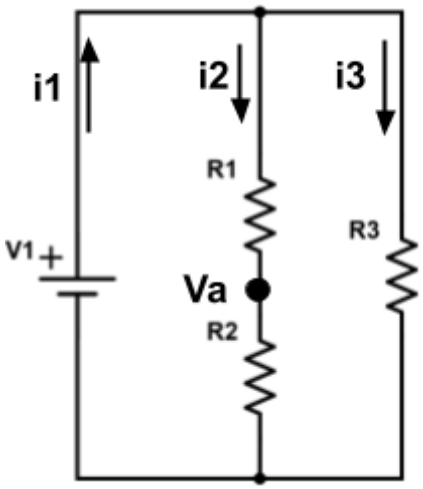
i1 = **i2=** **i3=** **Va=** **Vb=**

24) Dado el siguiente circuito, R1=R2=2Ω , R3=6Ω, V1=70V y V2=35V,encuentre los valores de **i1, i2 e i3 y la tensión en Va**.

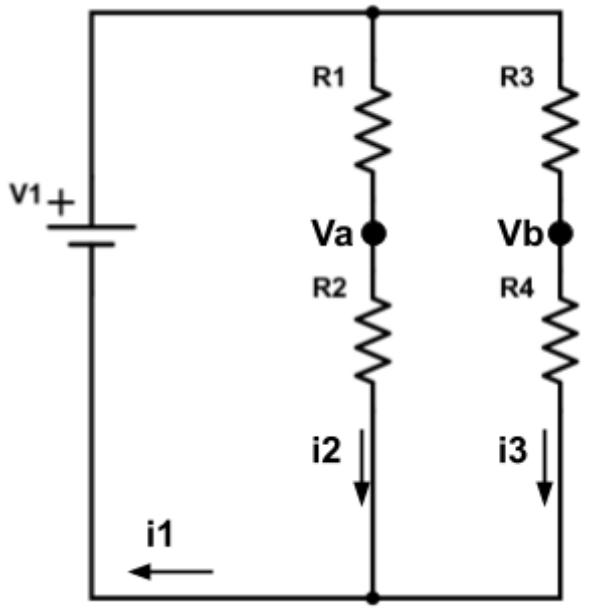


i1 = **i2=** **i3 =** **Va=**

25) Dado el siguiente circuito , se sabe que V1=100V , Va=25V , R3 = 200 Ω, i1=1,75A Encuentre I2,R1 y R2

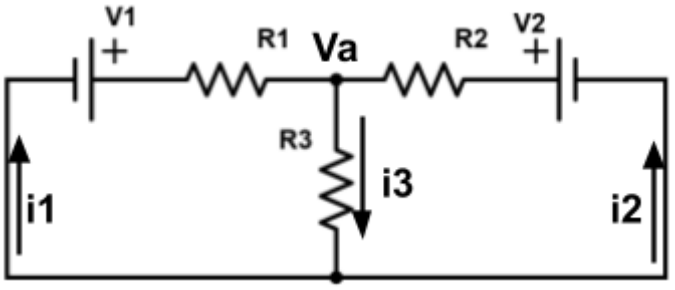


26) Dado el siguiente circuito, se sabe V1=30V, R1=6 Ω , R2=9 Ω , R3=5 Ω , R4=0 Ω. Calcule los valores de **i1, i2 e i3,** y las tensiones en **Va y Vb** con respecto al negativo de la fuente.



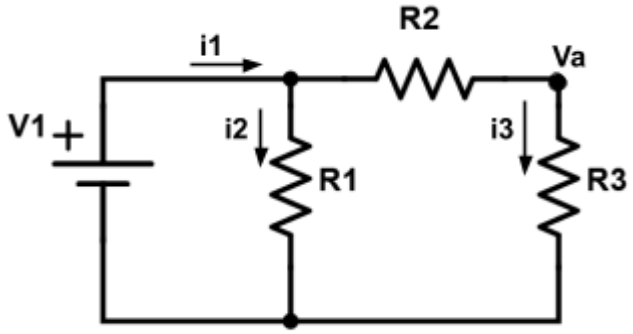
i1 = **i2=** **i3=** **Va=** **Vb=**

27) Dado el siguiente circuito, R1=R2=2Ω , R3=6Ω, V1=35V y V2=70V, encuentre los valores de **i1, i2 e i3 y la tensión en Va**

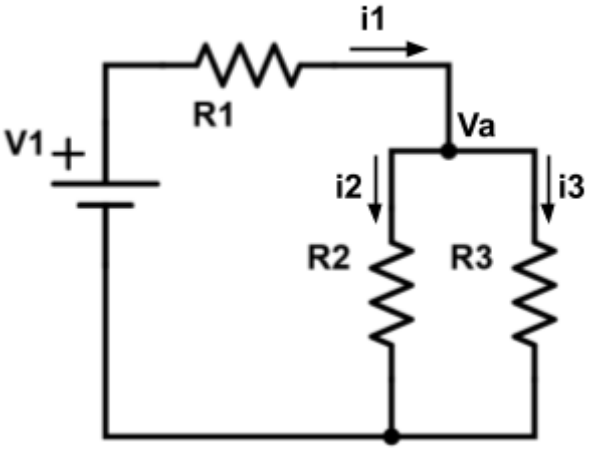


i1 = **i2=** **i3 =** **Va=**

28) Dado el siguiente circuito, se sabe V1 =12V, Va = 4,5V, i2 = 1mA, i3 = 3mA . Calcule R1,R2 y R3

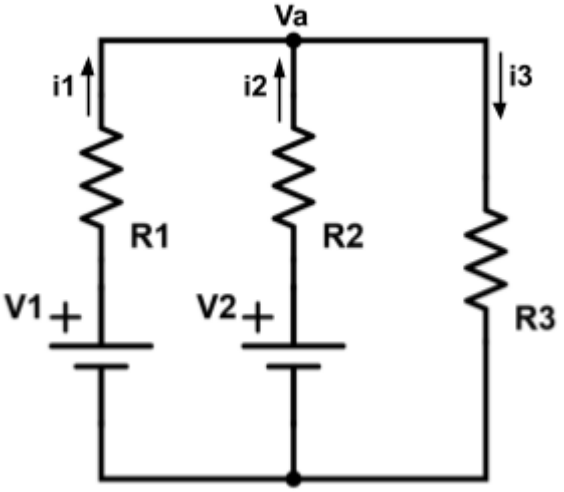


29) Dado el siguiente circuito, sabiendo que V1= 18V, Va = 5V, i2 = 5mA, R3 = 625Ω. Calcule:



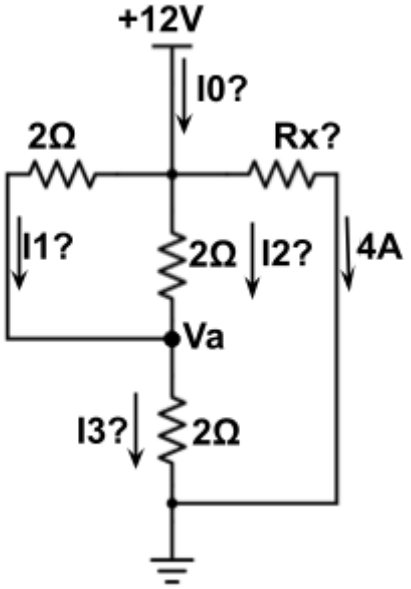
I1= **R1=** **R2=** **I3=**

30) Dado el siguiente circuito, sabiendo que V1= 40V, V2 = 20V, R1 = 10Ω, R2=10Ω, R3 = 15Ω. Calcule:



I1= **I2=** **I3=** **Va**

31)



I0=

I1=

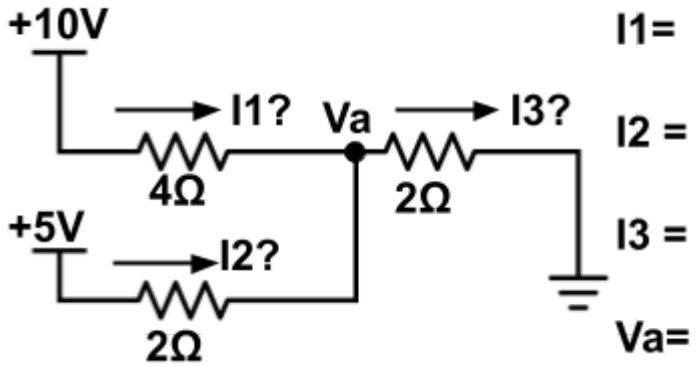
I2=

I3=

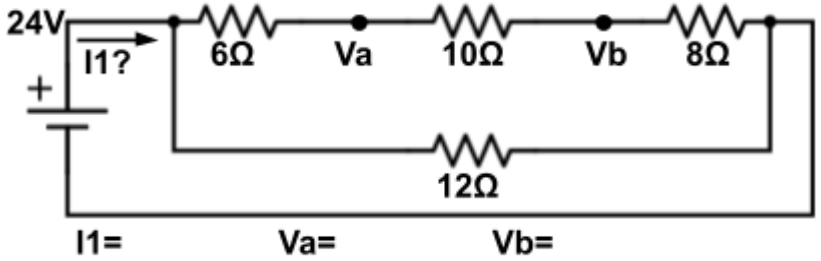
Va=

Rx=

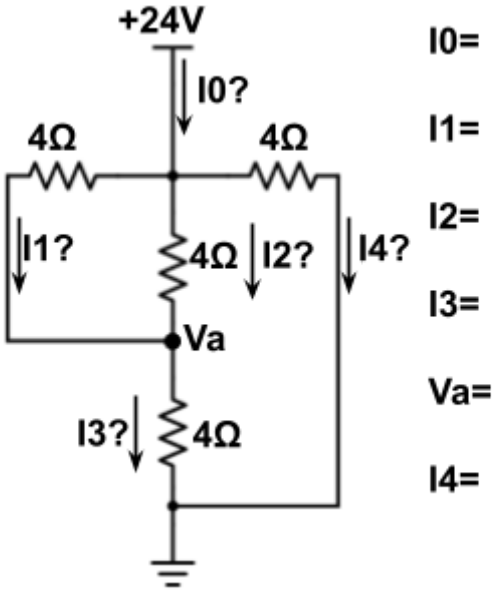
32)


+10V
I1?
Va
I3?
4Ω
2Ω
+5V
I2?
2Ω
I1=
I2 =
I3 =
Va=

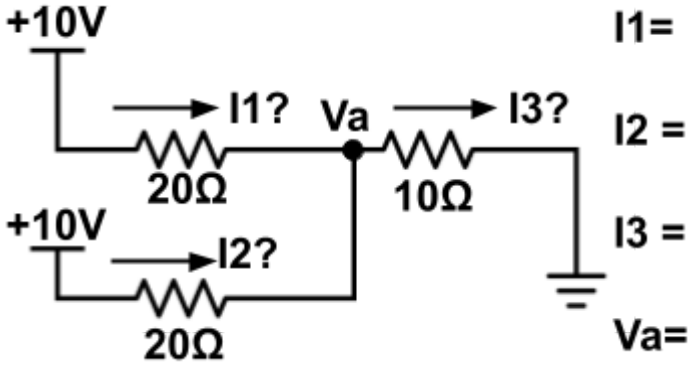
33)


24V
I1?
6Ω
Va
10Ω
Vb
8Ω
12Ω
I1=
Va=
Vb=

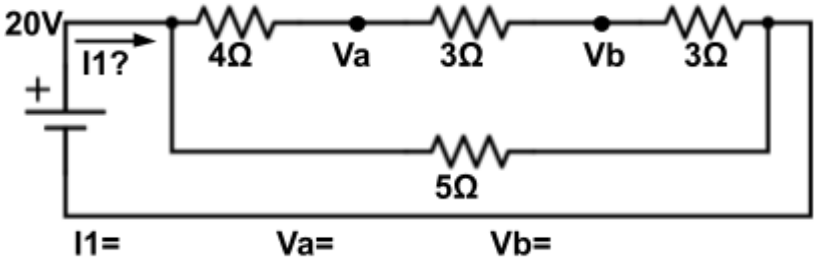
34)


+24V
I0?
4Ω
4Ω
I1?
4Ω
I2?
I4?
Va
I3?
4Ω
I0=
I1=
I2=
I3=
Va=
I4=

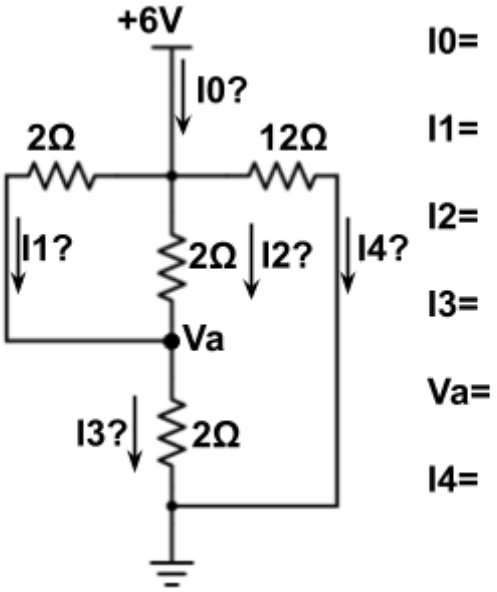
35)


+10V
I1?
Va
I3?
20Ω
10Ω
+10V
I2?
20Ω
I1=
I2 =
I3 =
Va=

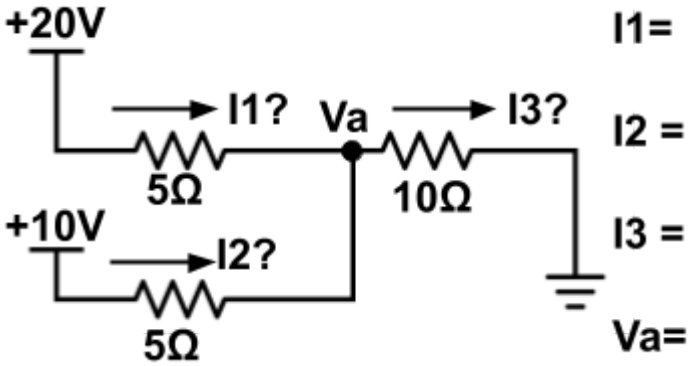
36)


20V
+
I1?
4Ω
Va
3Ω
Vb
3Ω
5Ω
I1=
Va=
Vb=

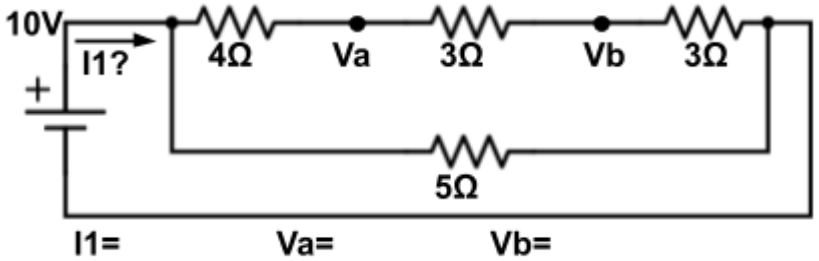
37)


+6V
I0?
2Ω
12Ω
I1?
2Ω
I2?
I4?
Va
I3?
2Ω
I0=
I1=
I2=
I3=
Va=
I4=

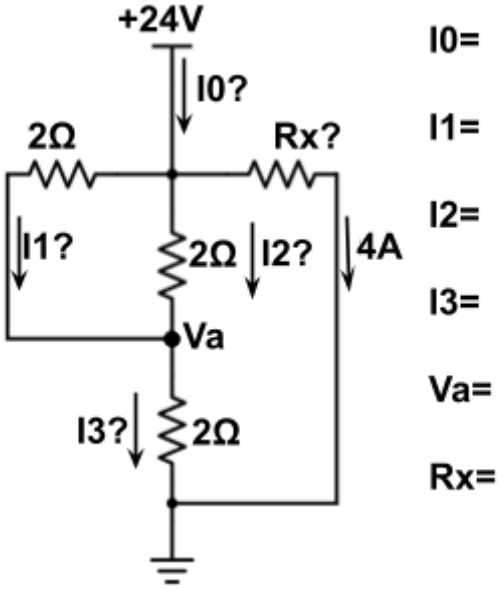
38)


+20V
I1?
Va
I3?
5Ω
10Ω
+10V
I2?
5Ω
I1=
I2 =
I3 =
Va=

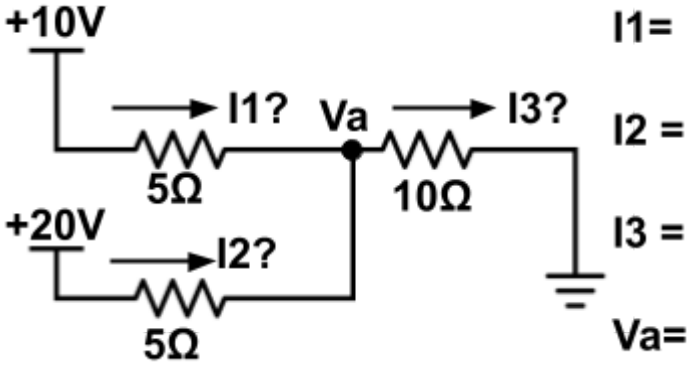
39)


10V
+
I1?
4Ω
Va
3Ω
Vb
3Ω
5Ω
I1=
Va=
Vb=

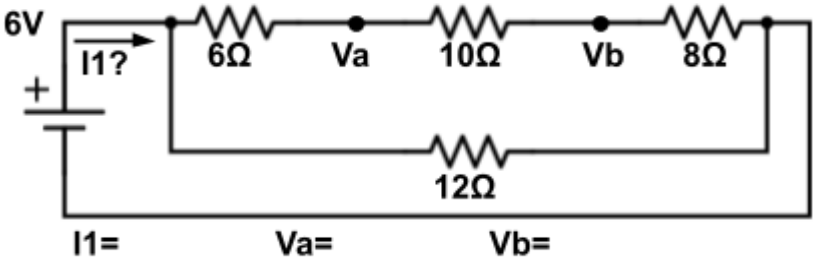
40)


+24V
I0?
2Ω
Rx?
I1?
2Ω
I2?
4A
Va
I3?
2Ω
I0=
I1=
I2=
I3=
Va=
Rx=

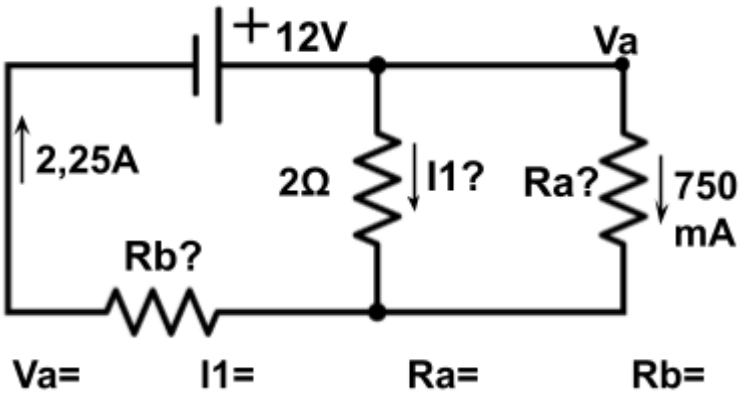
41)


+10V
I1?
Va
I3?
5Ω
10Ω
+20V
I2?
5Ω
I1=
I2 =
I3 =
Va=

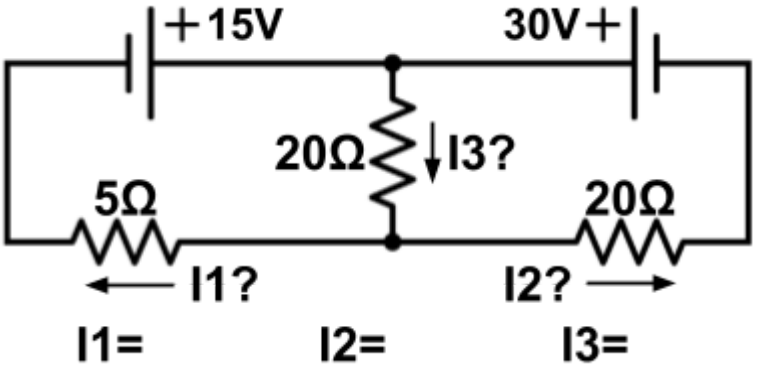
42)


6V
I1?
+
6Ω
Va
10Ω
Vb
8Ω
12Ω
I1=
Va=
Vb=

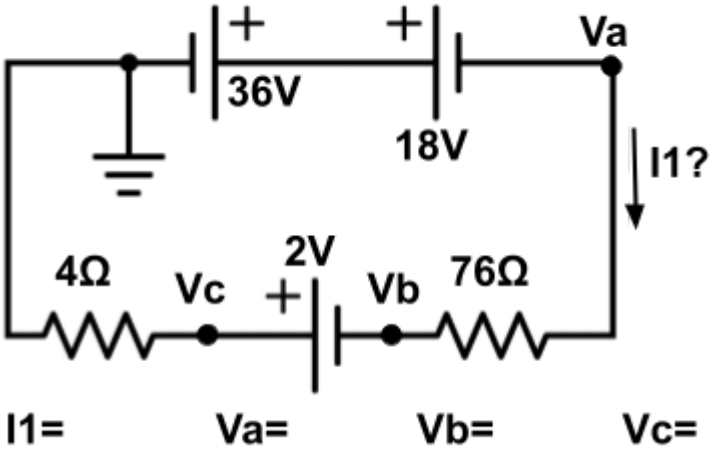
43)


+12V
Va
2,25A
2Ω
I1?
Ra?
750 mA
Rb?
Va=
I1=
Ra=
Rb=

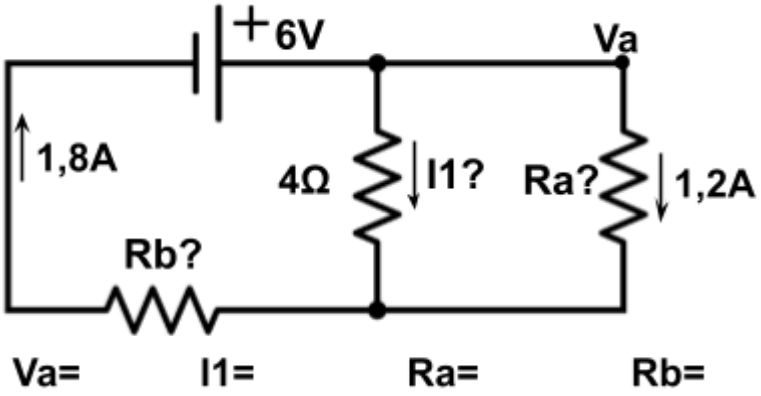
44)


+15V
30V+
20Ω
I3?
5Ω
20Ω
I1?
I2?
I1=
I2=
I3=

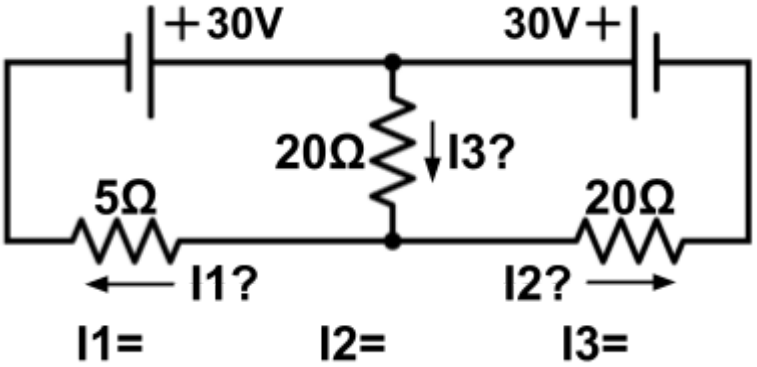
45)


+
+
Va
36V
18V
I1?
2V
4Ω
Vc
+
Vb
76Ω
I1=
Va=
Vb=
Vc=

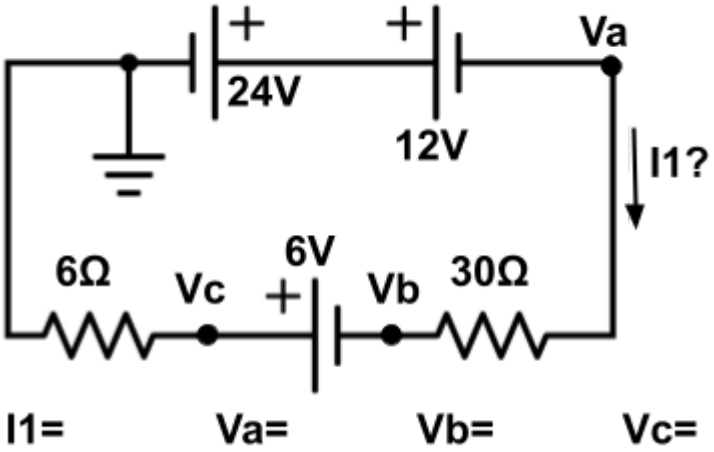
46)


+6V
Va
1,8A
4Ω
I1?
Ra?
1,2A
Rb?
Va=
I1=
Ra=
Rb=

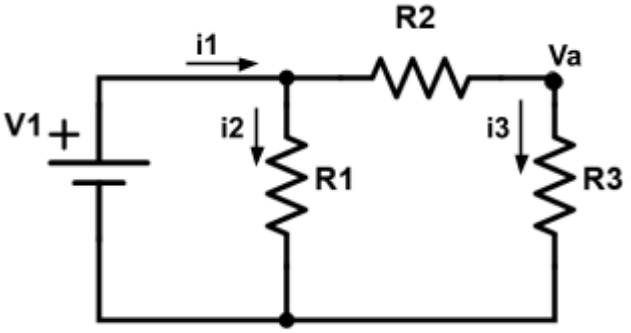
47)



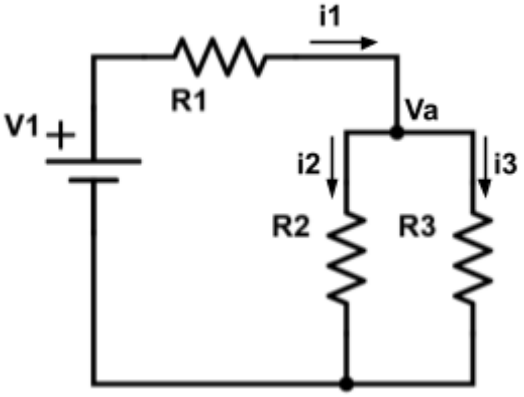
48)



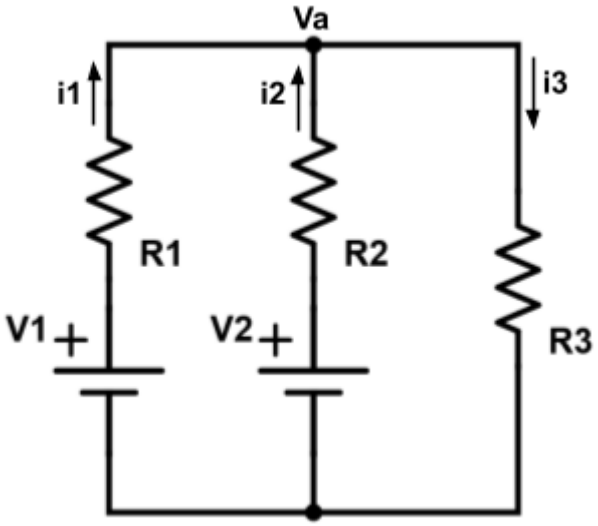
49) Dado el siguiente circuito, se sabe V1 =36V, R1 = 3kΩ, R2 = 1k1Ω, R3 = 700Ω . Calcule I2,I3,Va



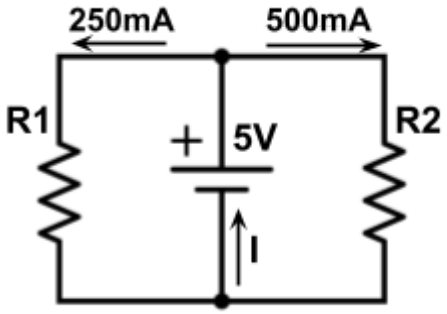
50) Dado el siguiente circuito, sabiendo que V1= 15V, Va = 6V, R2 = 2k4Ω, i3 = 2 mA. Calcule: I1,R1,R3,I2



51) Dado el siguiente circuito, sabiendo que V1= 10V, V2 = 30V, R1 = 5Ω, R2=5Ω, R3 = 10Ω. Calcule: I1,I2,I3,Va



52)

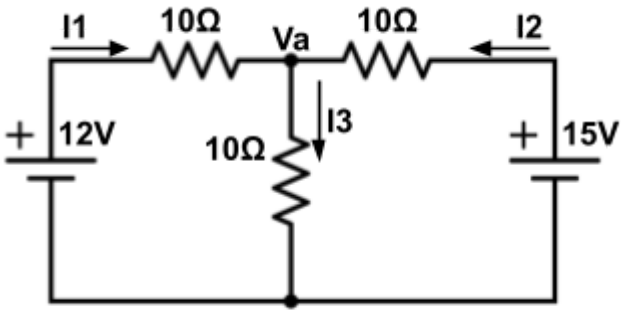


I =

R1 =

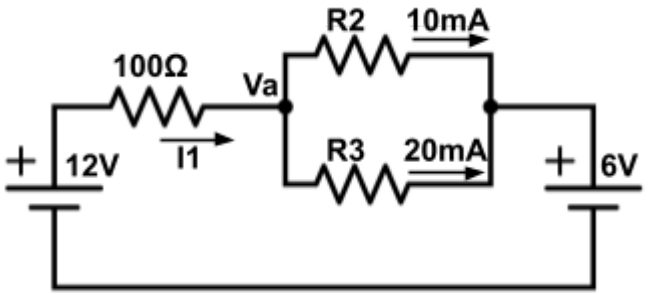
R2 =

53)



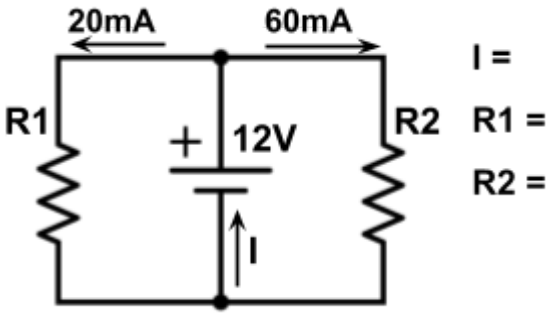
I1 = I2 = I3 = Va=

54)



I1= Va= R2= R3=

55)

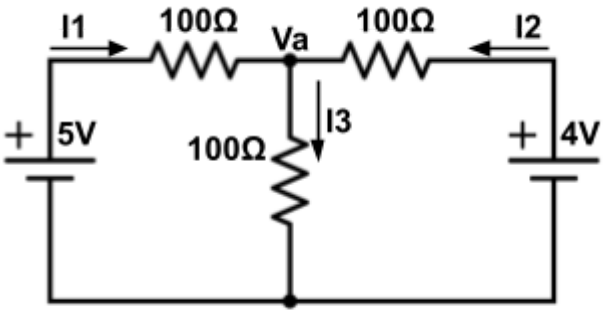


I =

R1 =

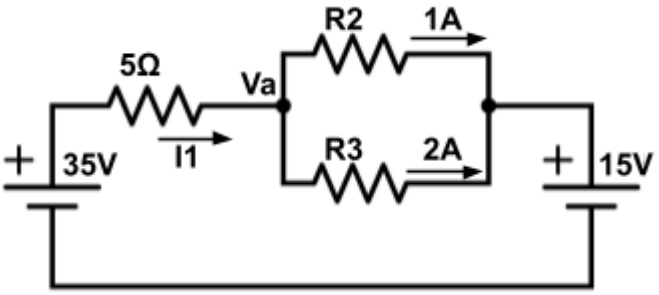
R2 =

56)



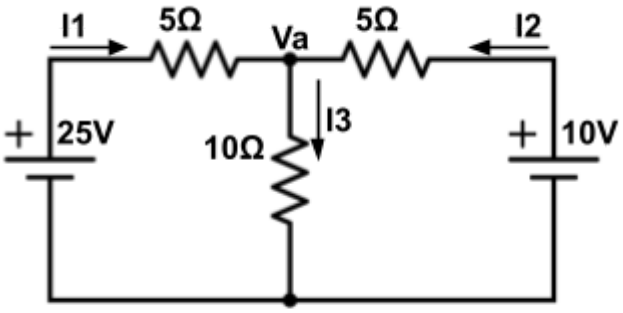
I1 = I2 = I3 = Va=

57)



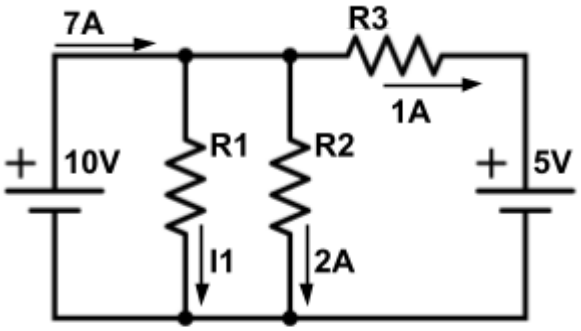
I1= Va= R2= R3=

58)



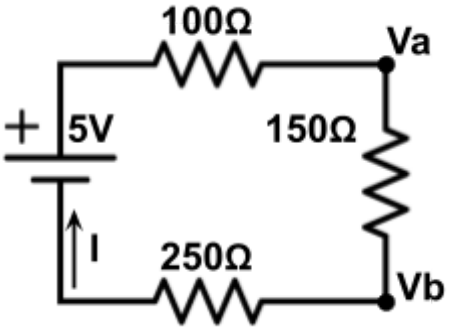
I1 = I2 = I3 = Va=

59)



I1= R1= R2= R3=

60)

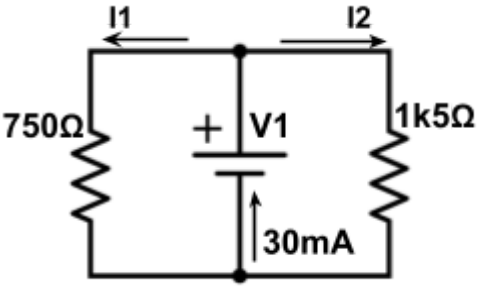


I =

Va =

Vb =

61)

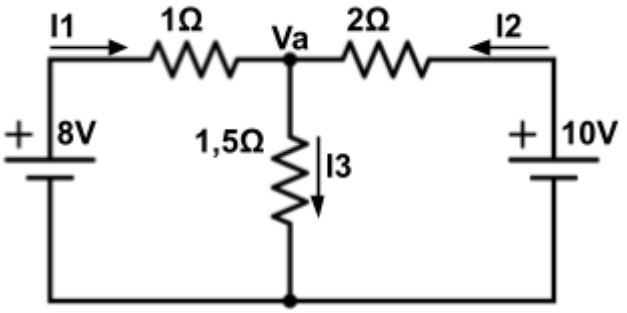


V1 =

I1 =

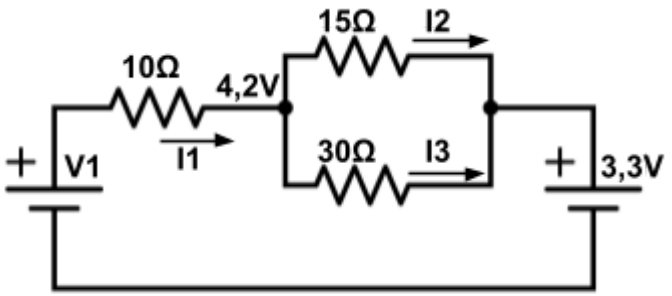
I2 =

62)



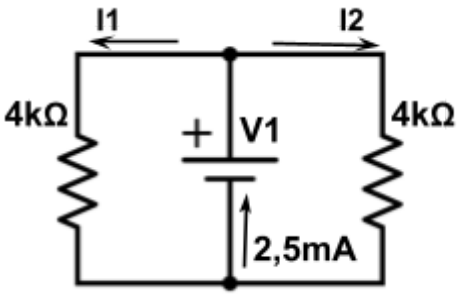
I1 = I2 = I3 = Va=

63)



I1= I2= I3= V1=

64)

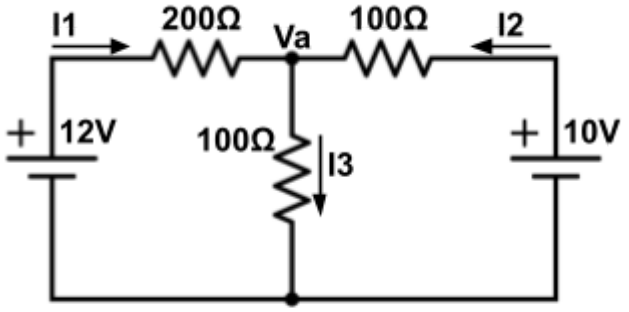


V1 =

I1 =

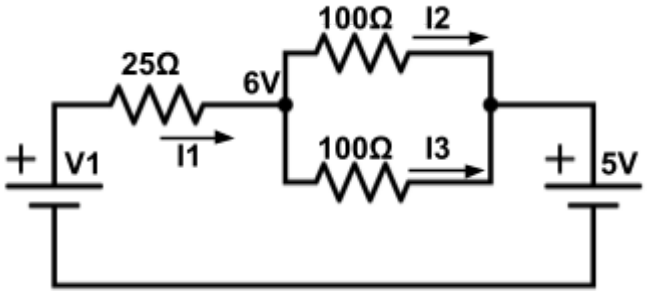
I2 =

65)



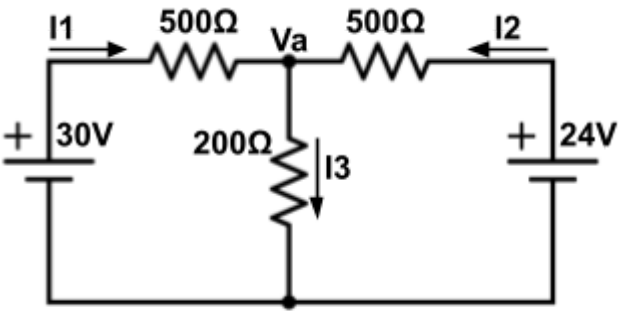
I1 = I2 = I3 = Va=

66)



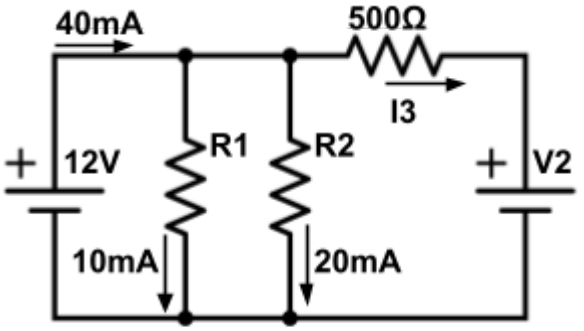
I1= I2= I3= V1=

67)



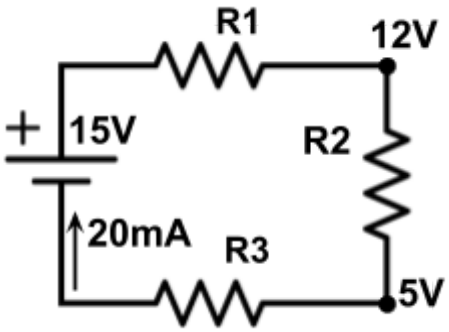
I1 = I2 = I3 = Va=

68)



I3= R1= R2= V2=

69)

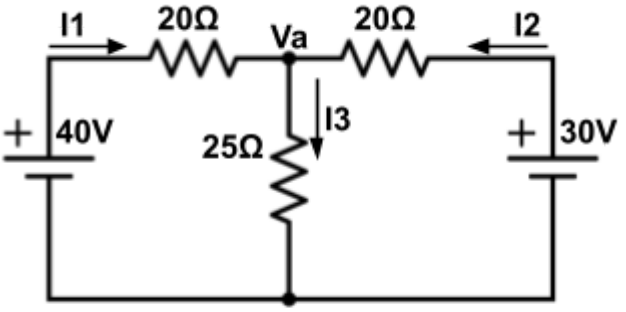


R1=

R2 =

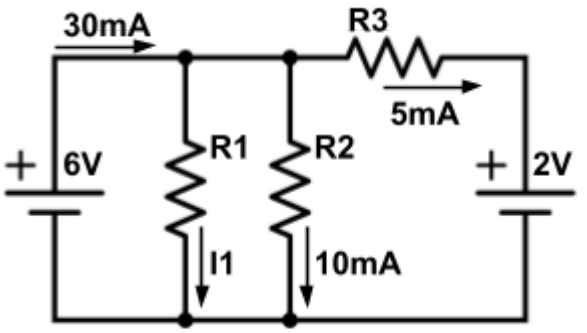
R3 =

70)



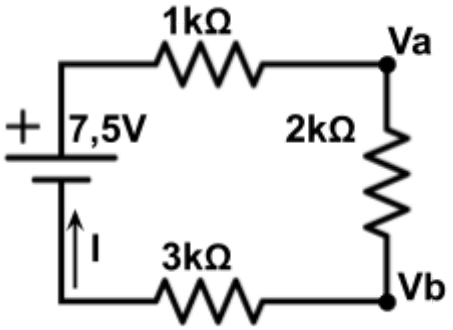
I1 = I2 = I3 = Va=

71)



I1= R1= R2= R3=

72)

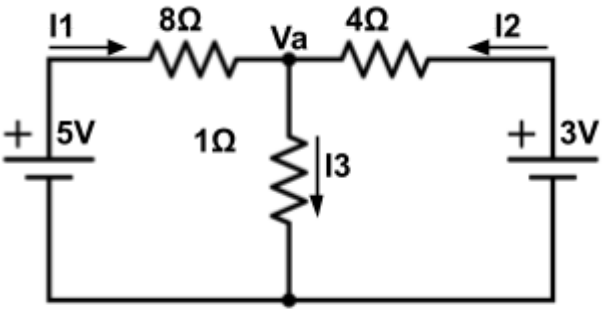


I =

Va =

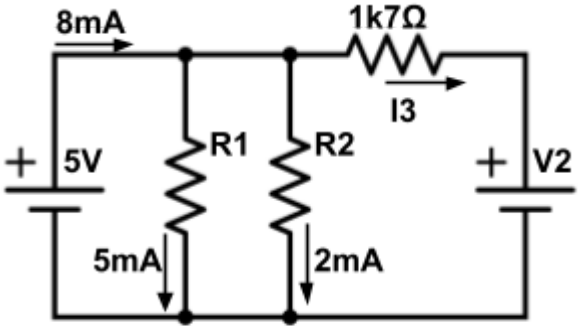
Vb =

73)



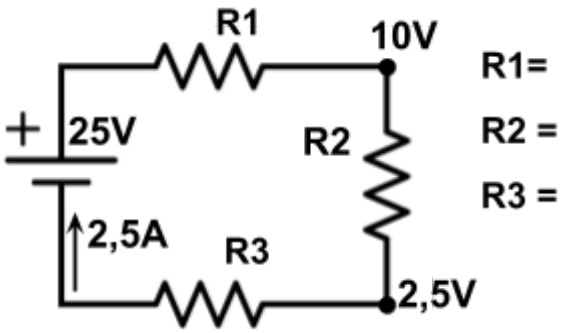
I1 = I2 = I3 = Va=

74)

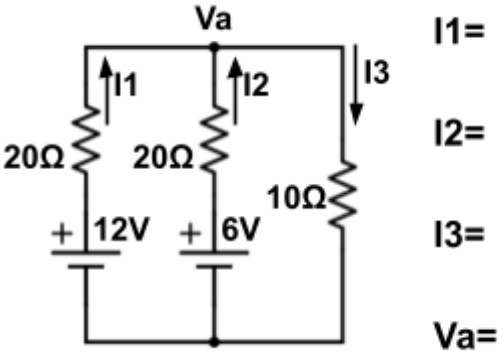


I3= R1= R2= V2=

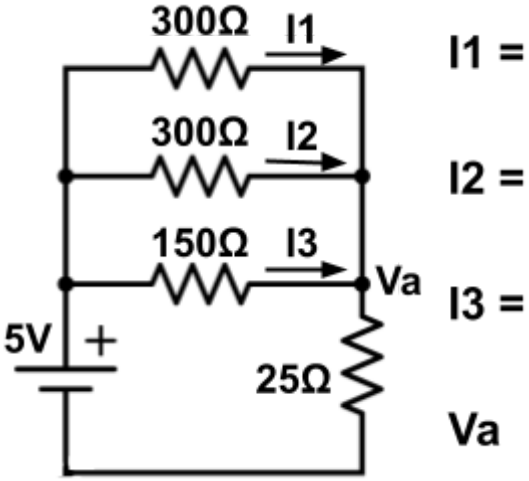
75)


R1
10V
+ 25V
R2
2,5A
R3
2,5V
R1=
R2 =
R3 =

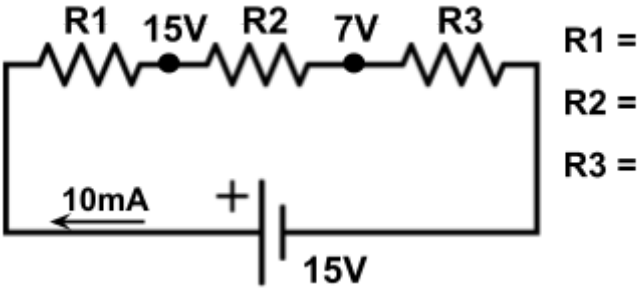
76)


Va
I1
I2
I3
20Ω
20Ω
10Ω
+ 12V
+ 6V
I1=
I2=
I3=
Va=

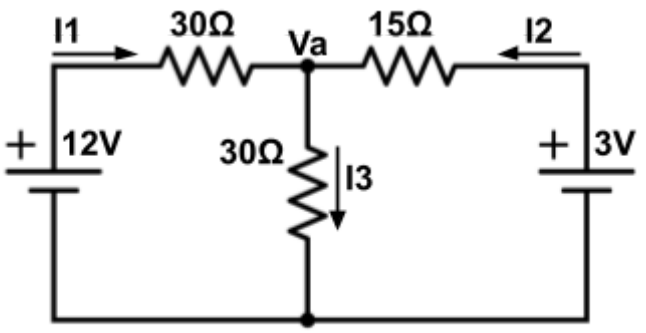
77)


300Ω
I1
300Ω
I2
150Ω
I3
Va
5V +
25Ω
I1 =
I2 =
I3 =
Va

78)

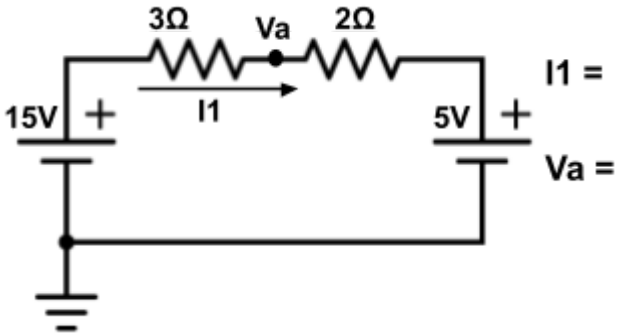

R1
15V
R2
7V
R3
10mA
+
15V
R1 =
R2 =
R3 =

79)



I1 = I2 = I3 = Va=

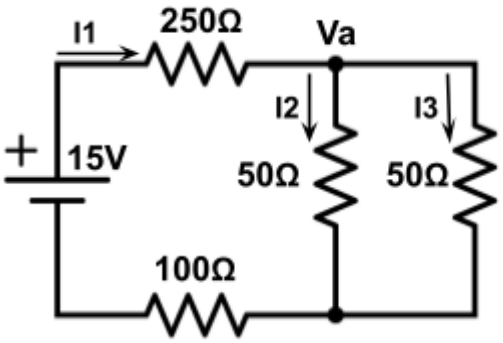
80)



I1 =

Va =

81)



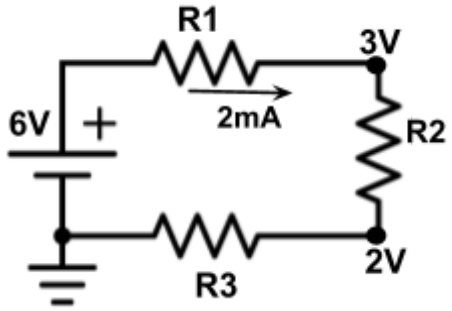
Va =

I1 =

I2 =

I3 =

82)

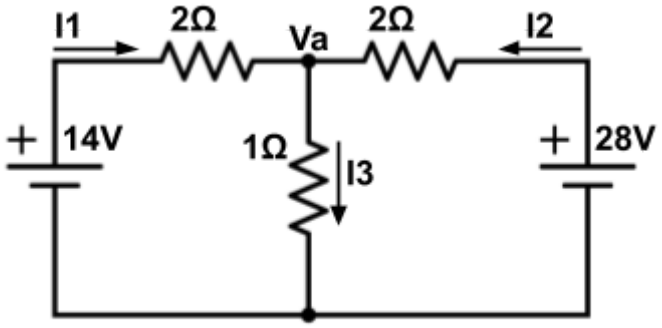


R1 =

R2 =

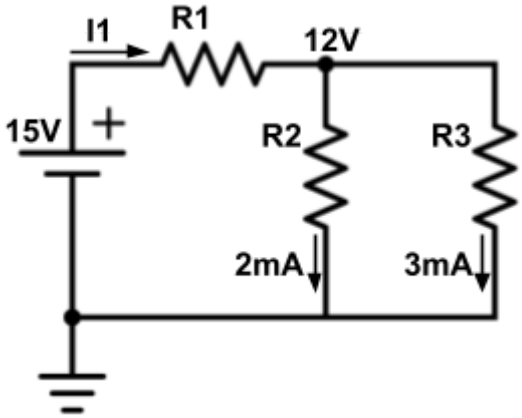
R3 =

83)



I1 = I2 = I3 = Va=

84)



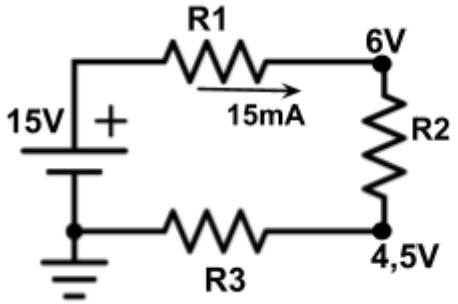
R1 =

R2 =

R3 =

I1=

85)

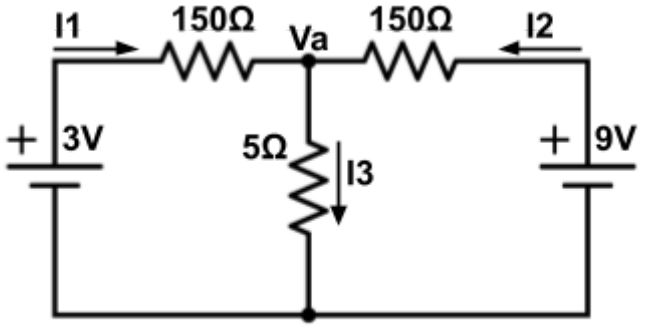


R1 =

R2 =

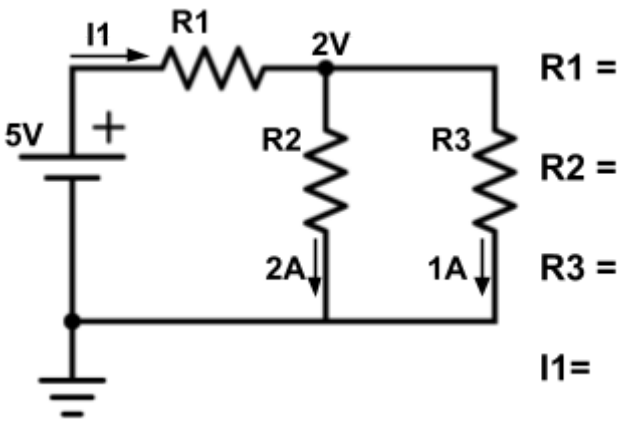
R3 =

86)

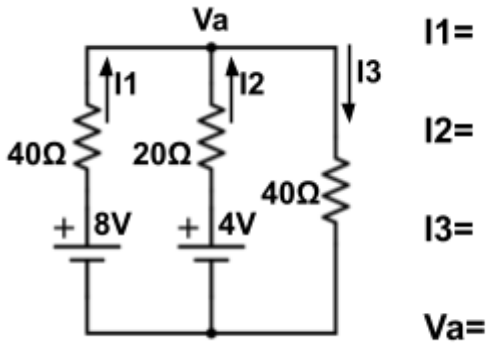


I1 = I2 = I3 = Va=

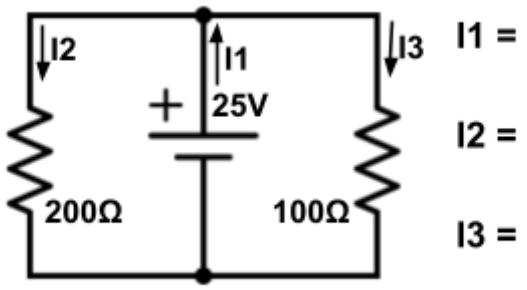
87)


I1
R1
2V
5V
+
R2
R3
2A
1A
R1 =
R2 =
R3 =
I1=

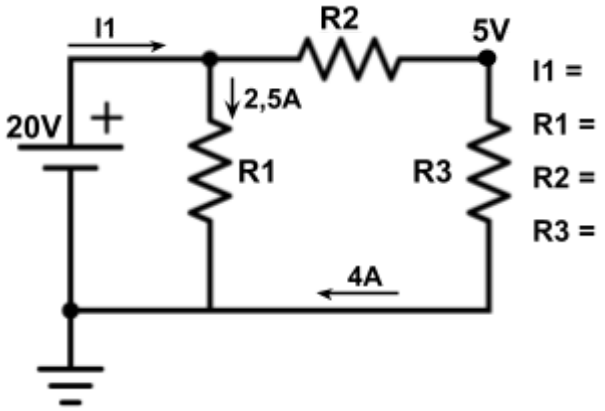
88)


Va
I1
I2
I3
40Ω
20Ω
40Ω
+ 8V
+ 4V
I1=
I2=
I3=
Va=

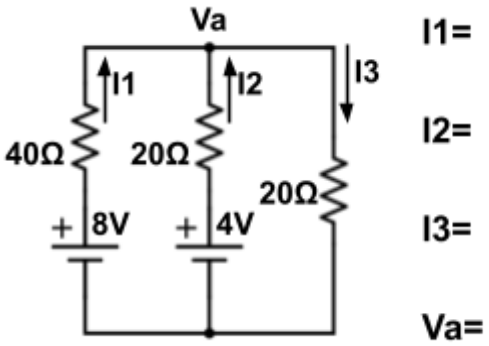
89)


I2
I1
I3
+
25V
200Ω
100Ω
I1 =
I2 =
I3 =

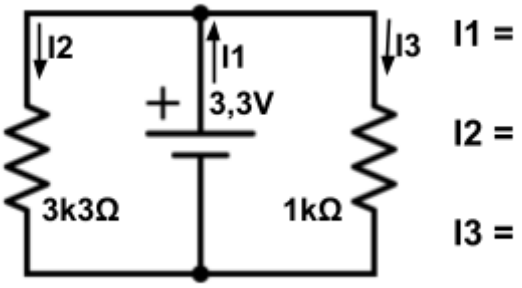
90)


I1
R2
5V
2,5A
20V
+
R1
R3
4A
I1 =
R1 =
R2 =
R3 =

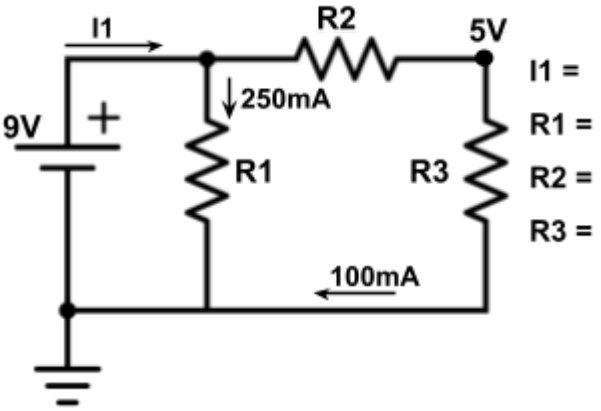
91)



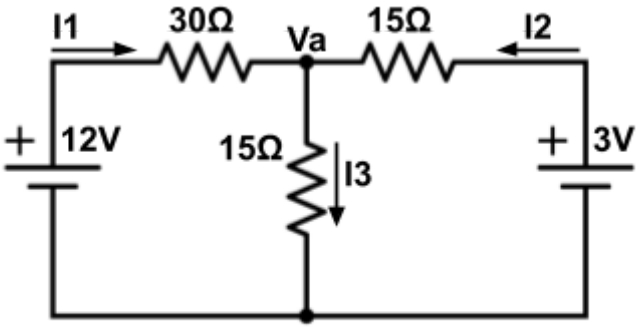
92)



93)

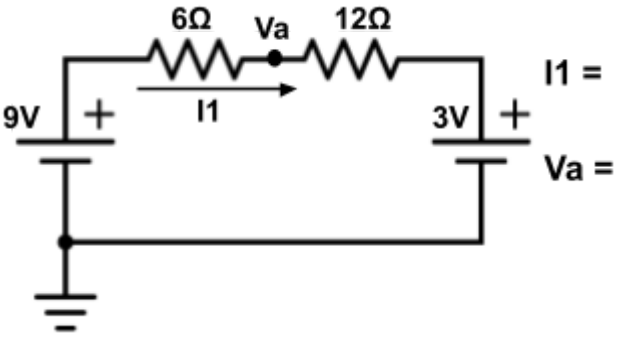


94)

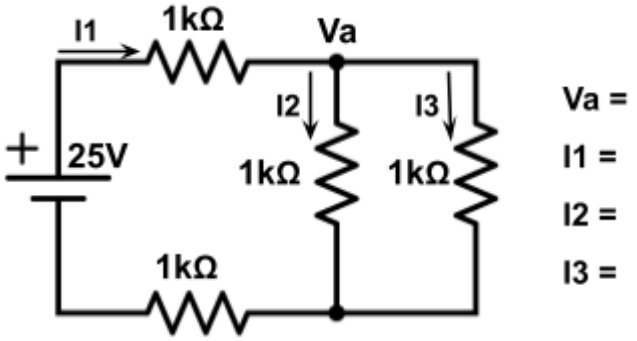


I1 = I2 = I3 = Va=

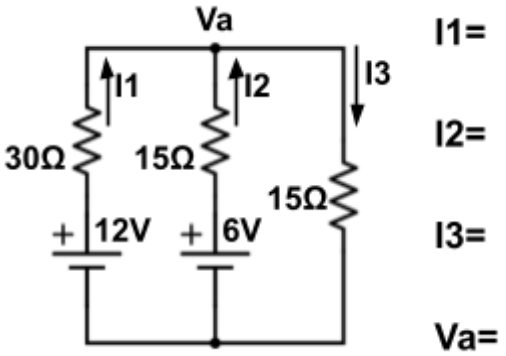
95)


6Ω
Va
12Ω
9V
+
I1
3V
+
I1 =
Va =

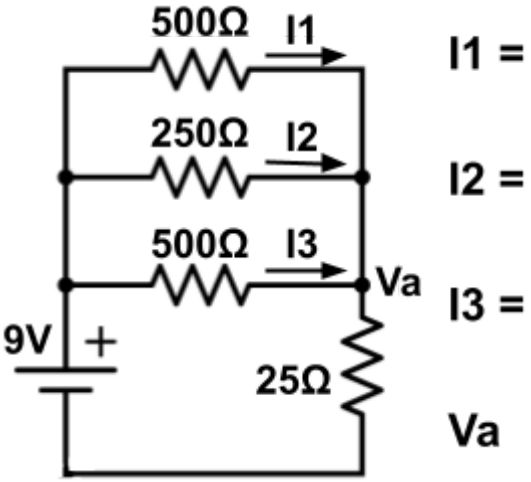
96)


I1
1kΩ
Va
I2
I3
+
25V
1kΩ
1kΩ
1kΩ
Va =
I1 =
I2 =
I3 =

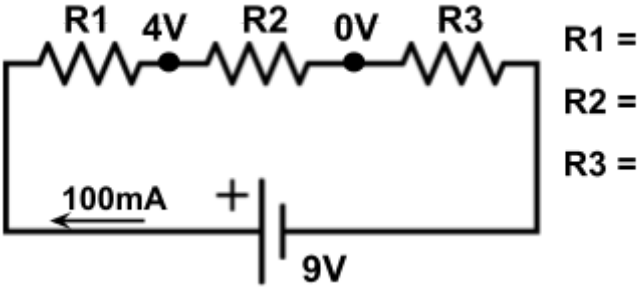
97)


Va
I1
I2
I3
30Ω
15Ω
15Ω
+
12V
+
6V
I1=
I2=
I3=
Va=

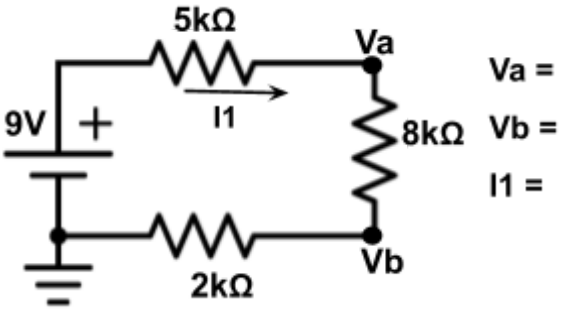
98)


500Ω
I1
250Ω
I2
500Ω
I3
Va
9V
+
25Ω
I1 =
I2 =
I3 =
Va

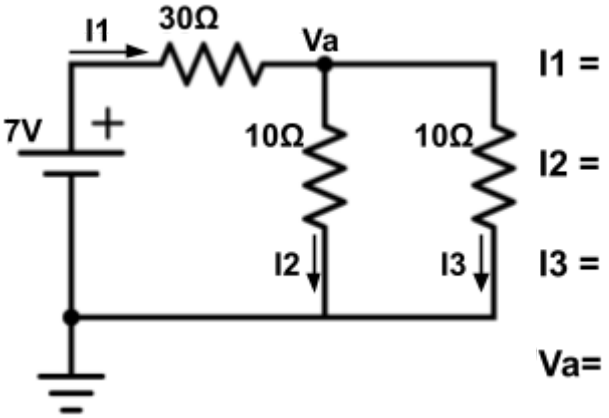
99)



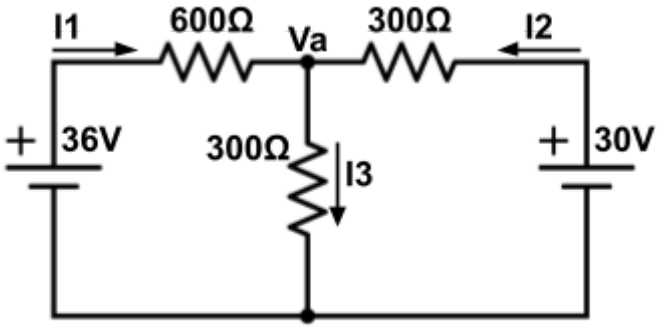
100)



101)

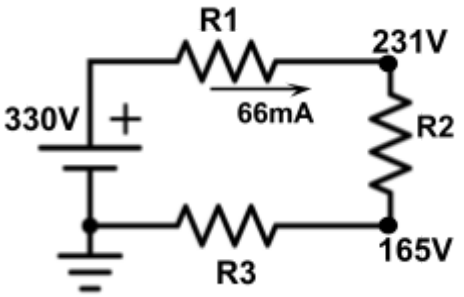


102)



I1 = I2 = I3 = Va=

103)

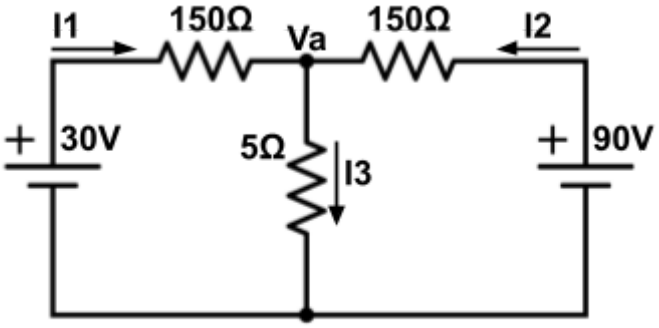


R1 =

R2 =

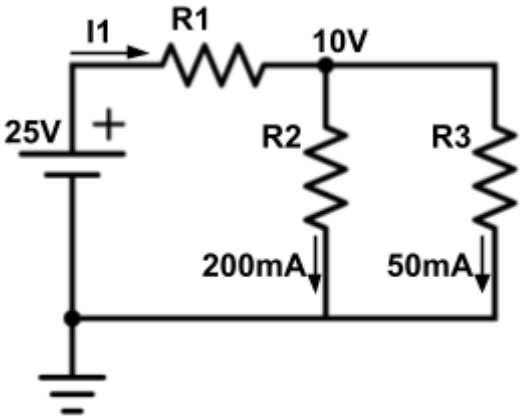
R3 =

104)



I1 = I2 = I3 = Va=

105)



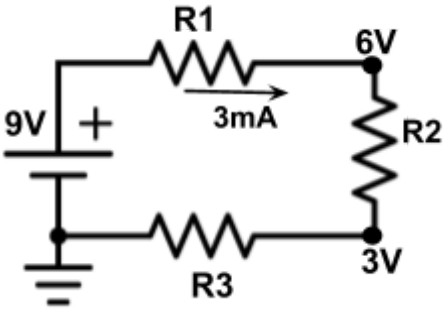
R1 =

R2 =

R3 =

I1=

106)

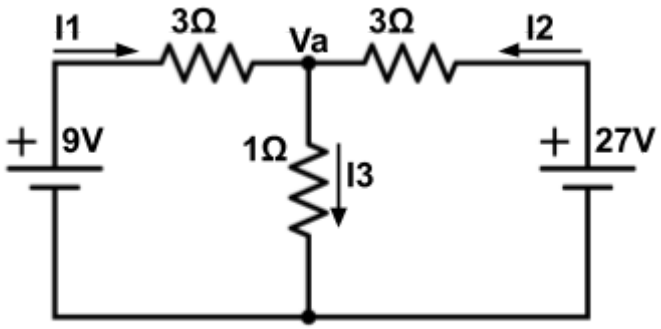


R1 =

R2 =

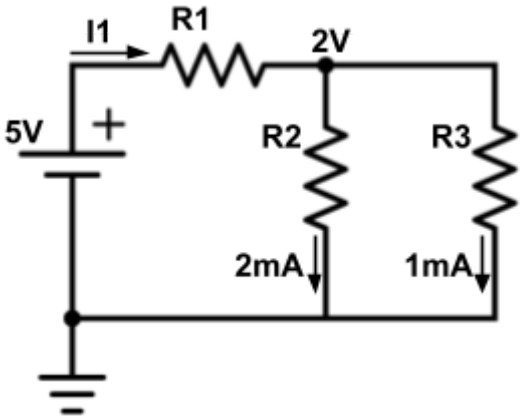
R3 =

107)



I1 = I2 = I3 = Va=

108)



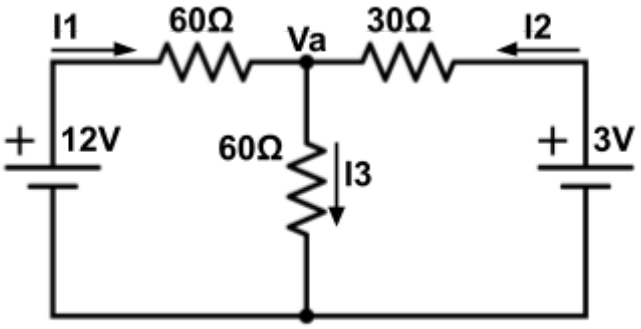
R1 =

R2 =

R3 =

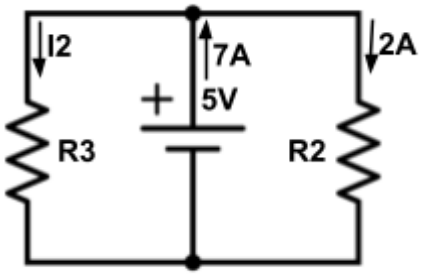
I1=

109)



I1 = I2 = I3 = Va=

110)

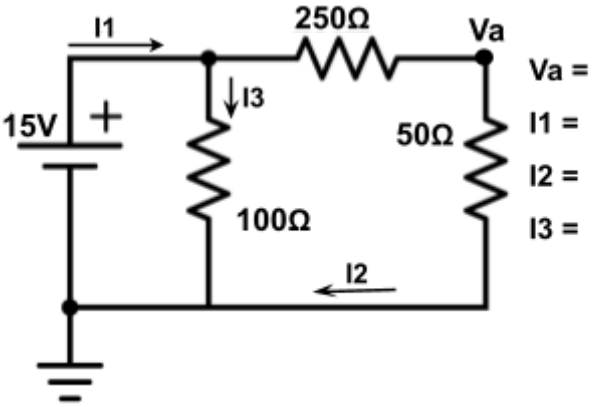


I2 =

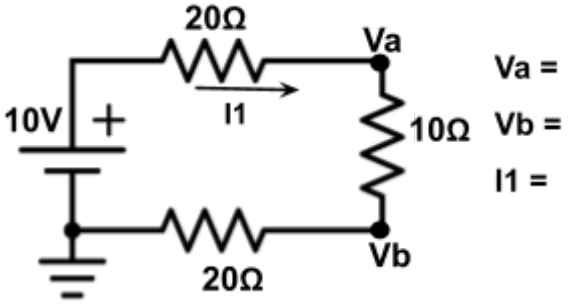
R1 =

R2 =

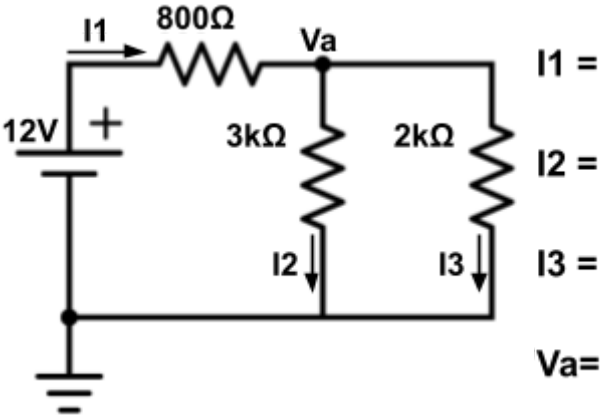
111)



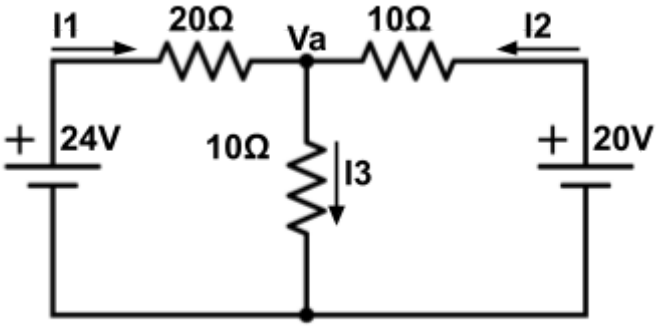
112)



113)

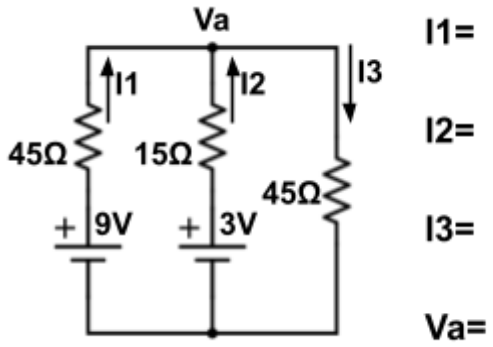


114)

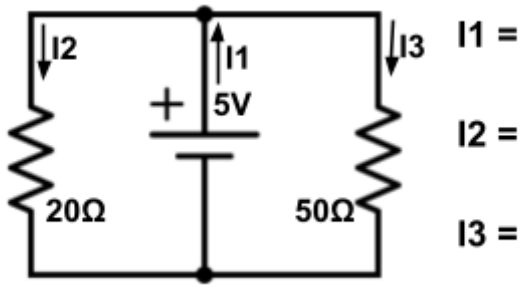


I1 = I2 = I3 = Va=

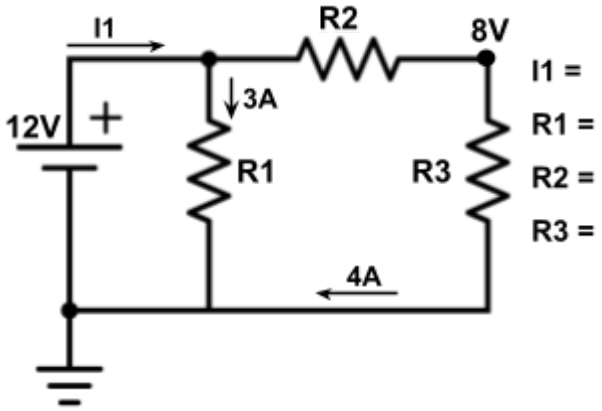
115)



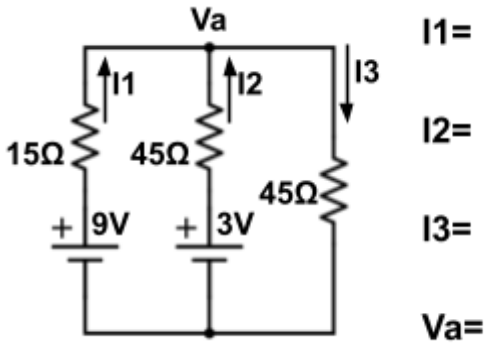
116)



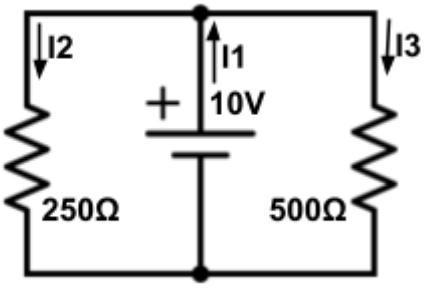
117)



118)



119)

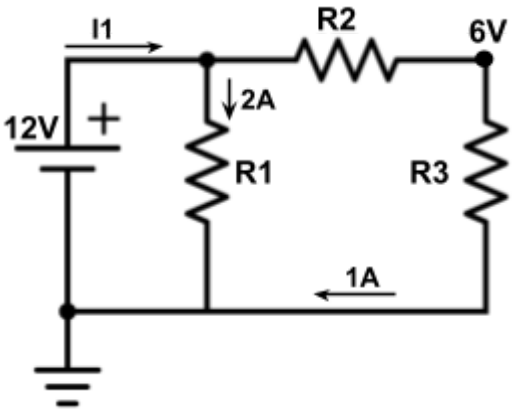


I1 =

I2 =

I3 =

120)



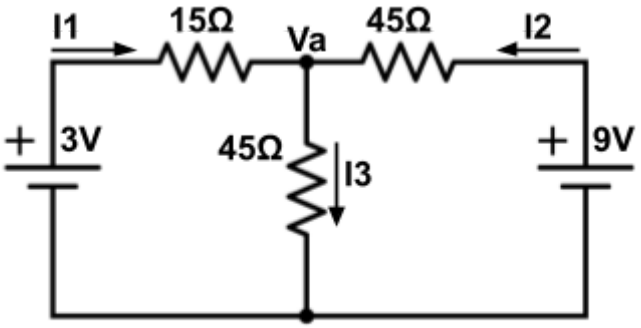
I1 =

R1 =

R2 =

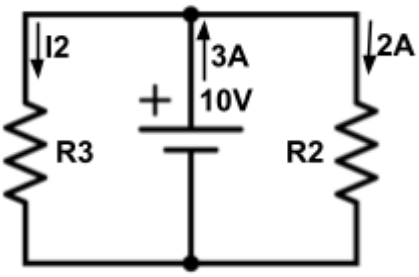
R3 =

121)



I1 = I2 = I3 = Va=

122)

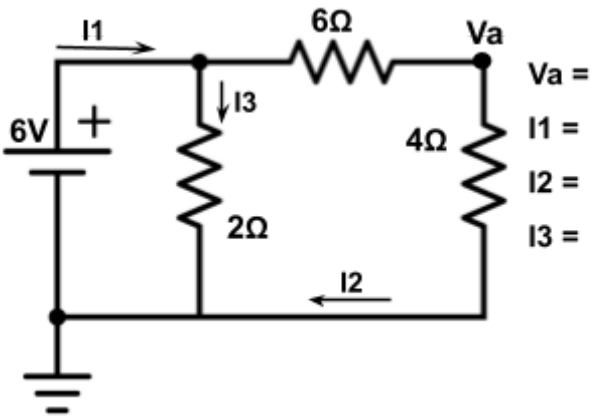


I2 =

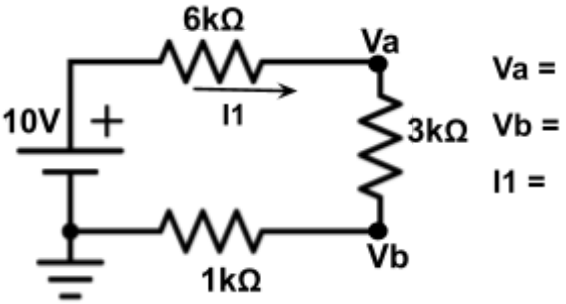
R1 =

R2 =

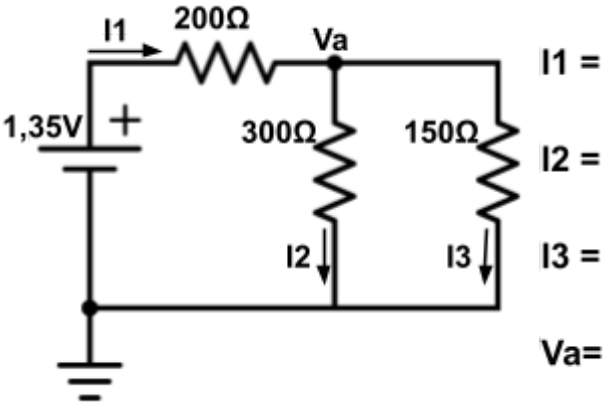
123)



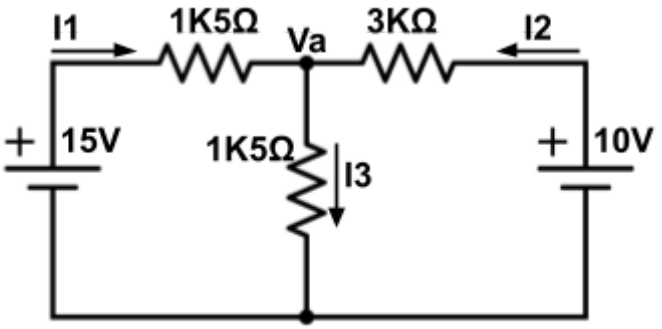
124)



125)

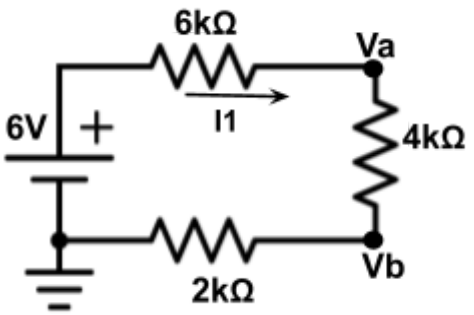


126)



I1 = I2 = I3 = Va=

127)

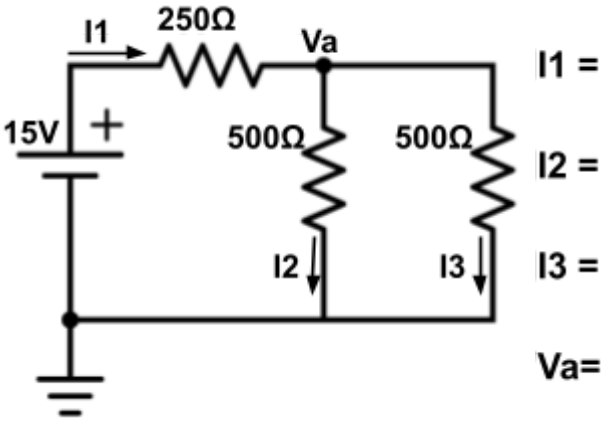


Va =

Vb =

I1 =

128)



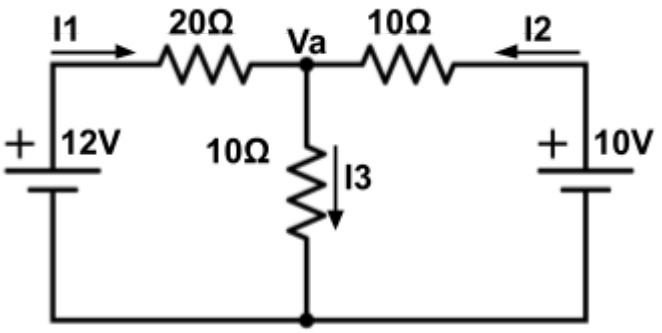
I1 =

I2 =

I3 =

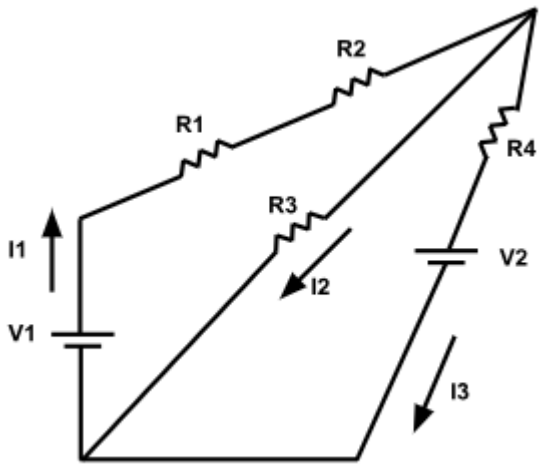
Va=

129)



I1 = I2 = I3 = Va=

130)



V1	13V
V2	1V
R1	15Ω
R2	5Ω
R3	
R4	10Ω
I1	500 mA
I2	
I3	